

Solving a system means finding the values of the variables. One way to solve a system is by graphing the equations.

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Solve: $\begin{cases} y = -x + 3 \\ y = \frac{1}{2}x - 3 \end{cases}$

Graph both equations on the same grid.

Remember, you can always graph a line by making a table of values and then plotting points.

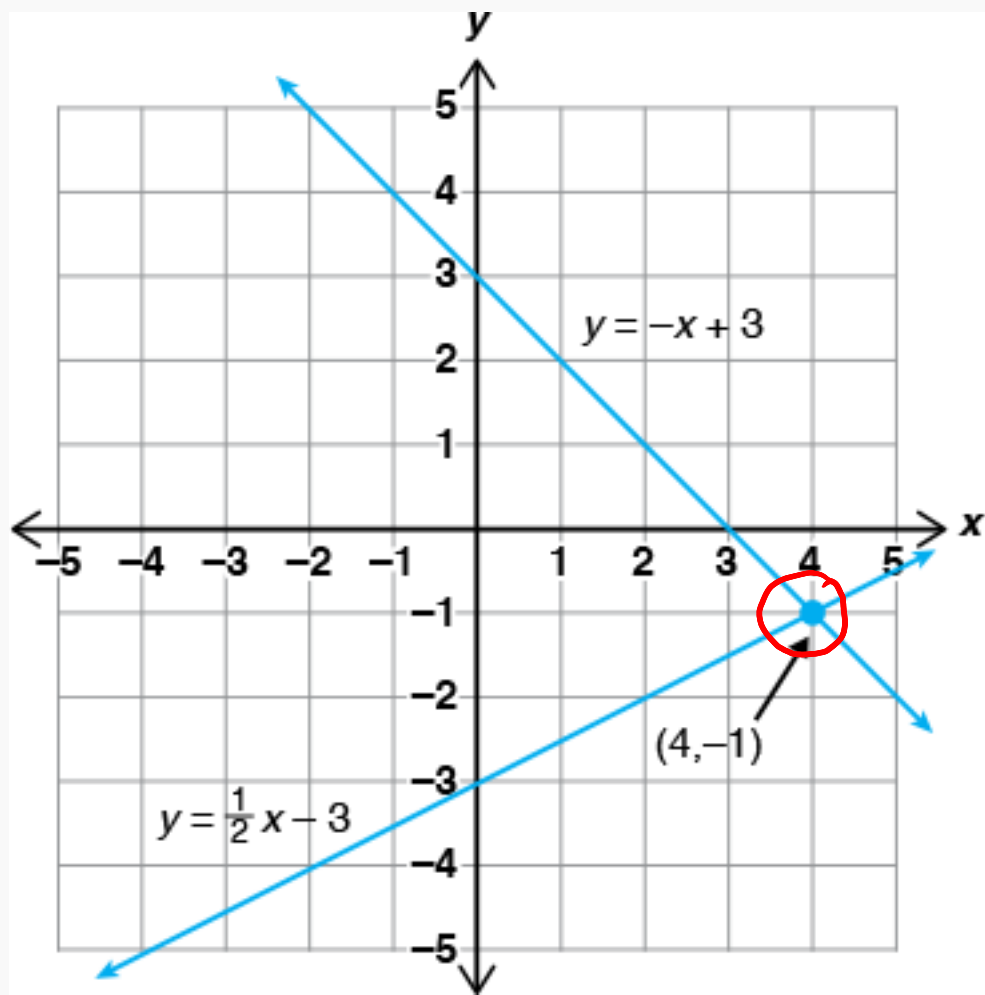
$$y = -x + 3$$

x	y
-1	4
0	3
1	2

$$y = \frac{1}{2}x - 3$$

x	y
-2	-4
0	-3
2	-2

The lines intersect at point $(4, -1)$. The x - and y -coordinates of this point are the solution to the system: $x = 4$ and $y = -1$.



Check your answer. Substitute the values into the system. Make sure both equations are true.

$$y = -x + 3$$
$$-1 = -(4) + 3$$
$$-1 = -1 \text{ True}$$

$$\begin{aligned} y &= \frac{1}{2}x - 3 \rightarrow y = \frac{1}{2}x - 3 \\ -1 &= \frac{1}{2}(4) - 3 \\ -1 &= 2 - 3 \\ -1 &= -1 \text{ True} \end{aligned} \quad \left. \begin{aligned} LHS &= -1 \\ RHS &= \frac{1}{2}(4) - 3 \\ &= 2 - 3 \\ &= -1 \end{aligned} \right\} \begin{aligned} LHS &= RHS \checkmark \\ &(\text{true}) \end{aligned}$$

system of equations